MACHINE LEARNING IN PHYSICS
SIMULATION-BASED INFERENCE

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INTRODUCTION

Introduction: Statistical Inference

Consider the following recent *measurement* of the Hubble constant[1]

$$H_0 = 73.30 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$$

What is the statistical meaning of the above statement?

If this a Bayesian statement, the meaning is: There is a 0.68 probability that

$$H_0 \in (72.36, 74.44)$$

1. Adam G. Riess *et al.* 2022. A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km s^{-1} Mpc^{-1} Uncertainty from the Hubble Space Telescope and the SH0ES Team. 2022 *ApJL* **934** L7.

Introduction: Statistical Inference

If, however,

$$H_0 = 73.30 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$$

is a frequentist statement, the meaning is more involved.

The meaning is that $H_0 \in (72.36, 74.44)$ is a member of an ensemble of statements 68% of which are true.

$(72.36, 74.44)$ is an example of a confidence interval. If the observations are repeated (infinitely!) many times, it is *guaranteed* that *at least* 68% of the intervals obtained will include the true value of the Hubble constant, *whatever the true value might be*.

Introduction: Statistical Inference

The key quantity in any serious statistical inference is the statistical model

$$p(X\,|\theta)$$

where X denotes the potential observations and θ the parameters of the statistical model, e.g., the Hubble constant.

When observations *D* are entered into the statistical model, the latter becomes the likelihood function

$$p(D\,|\theta)$$

which is key for constructing confidence intervals and, more generally, confidence sets.

Introduction: Statistical Inference

- Unfortunately, statistical models can be intractable.
- However, if one has access to a high-fidelity simulator, it is possible to make inferences without *explicit* knowledge of the statistical model.
- Simulation-based inference (SBI)[1,2,3] are methods for making inferences from real data using a simulator.

1. Kyle Cranmer, Johann Brehmer, and Gille Loupes, https://doi.org/10.1073/pnas.1912789117;
2. Likelihood-Free Frequentist Inference: Confidence Sets with Correct Conditional Coverage Niccolo Dalmasso, Luca Masserano, David Zhao, Rafael Izbicki, Ann B. Lee, arXiv:2107.03920v6 [stat.ML] 6 Apr 2023.
3. A. Al Kadhim, HBP, O. F. Prosper, *Mach. Learn.: Sci. Technol.* **5** (2024) 015020.

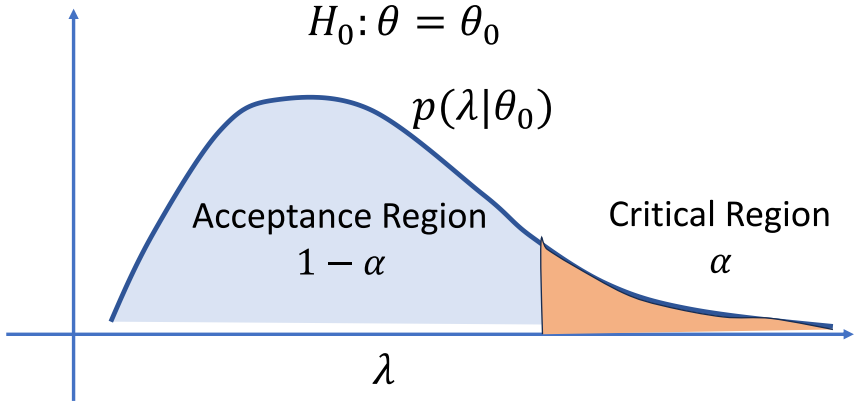
SIMULATION-BASED INFERENCE

SBI: Introduction

The key to SBI from the frequentist perspective is the relationship between hypothesis tests and confidence sets.

A hypothesis test $H_0: \theta = \theta_0$ typically proceeds as follows.

- Construct a function of potential observations X called a test statistic $\lambda(X, \theta)$ with the property that large values of λ cast doubt on the hypothesis H_0.

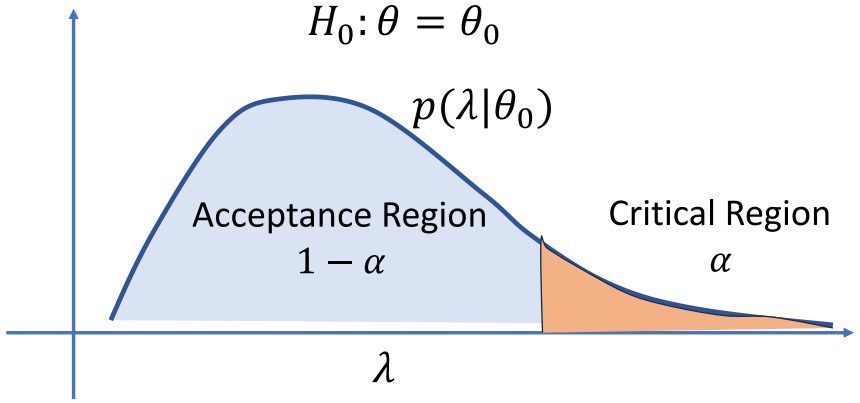


- Choose a small *fixed* probability α.
- Compute $p = P(\lambda > \lambda_D|\theta)$, where $\lambda_D = \lambda(D, \theta)$.
- If $p < \alpha$, reject the hypothesis, otherwise do *not* reject it.

SBI: Introduction

- For a given dataset $X = D$, repeated application of this test for different values of $\theta = \theta_0$ divides the parameter space into those θ_0 that are *rejected* and those that are *not*.
- The set of θ_0 that are *not* rejected is called a confidence set with associated confidence level $1 - \alpha$.
- Consequently, the criterion for *not* rejecting the hypothesis $H_0: \theta = \theta_0$, and, therefore, adding θ_0 to the confidence set is:

$$1 - P(\lambda > \lambda_D | \theta) \leq 1 - \alpha$$



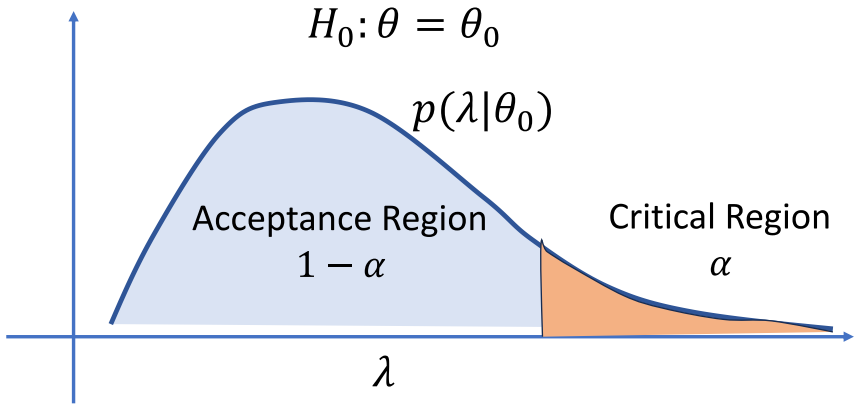
SBI: Introduction

- The key technical task is approximating either the *p-value*

$$p = P(\lambda > \lambda_D|\theta),$$

or, equivalently, the *cumulative density function* (cdf)

$$C(\lambda_D|\theta) \equiv 1 - p = P(\lambda \leq \lambda_D|\theta).$$



LIKELIHOOD FREE FREQUENTIST INFERENCE (LF2I)

LF2I

LF2I[1] is a suite of procedures for implementing frequentist-based inference when the likelihood function is intractable.

We'll focus on one aspect of LF2I, namely, the algorithm for approximating $\mathbb{C}(\lambda_D|\,\theta) \equiv P(\lambda \leq \lambda_D|\theta)$.

This algorithm requires:

1. A prior density $\pi(\theta)$ for sampling the parameter space.
2. A simulator $\mathcal{F}(\theta)$ to simulate potential observations, X.
3. A test statistic $\lambda(X, \theta)$.

1. Likelihood-Free Frequentist Inference: Confidence Sets with Correct Conditional Coverage Niccolo Dalmasso, Luca Masserano, David Zhao, Rafael Izbicki, Ann B. Lee, arXiv:2107.03920v6 [stat.ML] 6 Apr 2023.

LF2I: Algorithm

Algorithm 1 LF2I approximation of $\mathbb{C}(\lambda_D|\theta)$ given a simulator $\mathcal{F}(\theta)$

1. Initialize training sample $\mathbb{T} \leftarrow \varnothing$

while $k \in [1, \cdots K]$ **do**

2. Sample $\theta_k \sim \pi(\theta)$
3. Sample $X_k \equiv X_{1,k}, \cdots, X_{n,k} \sim \mathcal{F}(\theta_k)$
4. Compute test statistic $\lambda_k \leftarrow \lambda(X_k, \theta_k)$
5. Compute test statistic $\lambda_{D,k} \leftarrow \lambda(D, \theta_k)$
6. Compute indicator $Z_k \leftarrow \mathbb{I}(\lambda_k \leq \lambda_{D,k})$
7. Update training sample $\mathbb{T} \leftarrow \mathbb{T} \cup \{(\theta_k, Z_k)\}$

end while

8. Use $\mathbb{T}$ to train a machine learning (ML) model, $f(\theta; \omega)$, to approximate $\mathbb{C}(\lambda_D|\theta)$, where θ are the inputs to $f(\theta; \omega)$, and ω are the ML model parameters.

Neural network inputs: θ_k

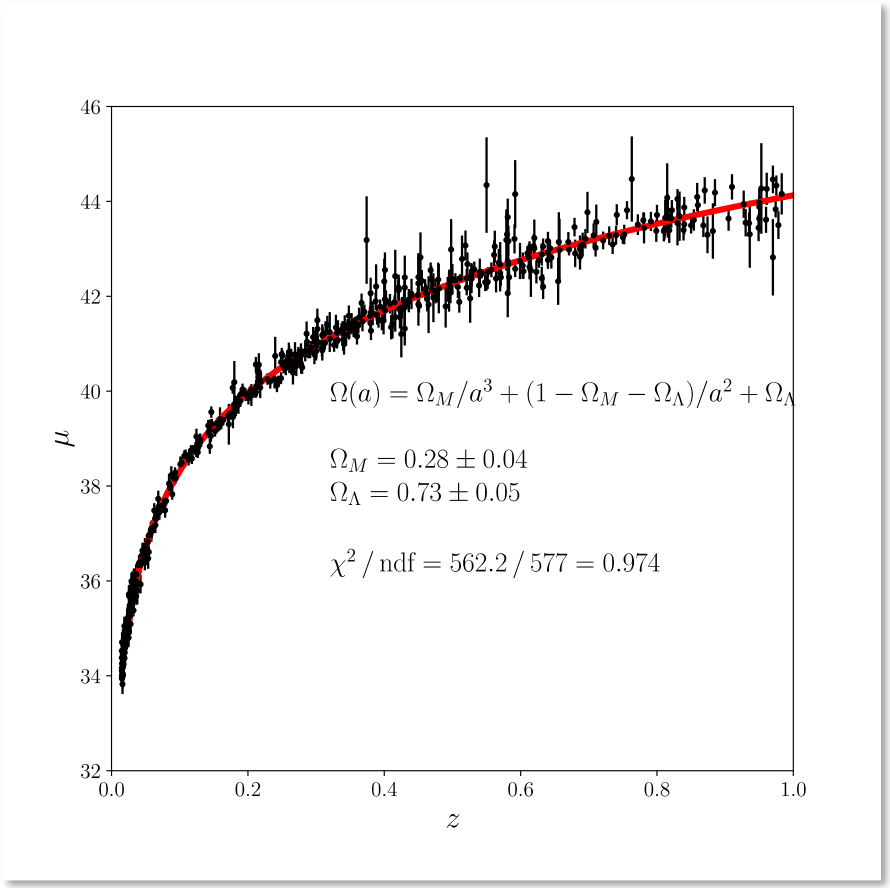
Targets: Z_k

Loss function: $(Z_k - f)^2$ $\qquad f \to \mathrm{E}[Z \mid \theta] = P(\lambda \leq \lambda_D|\theta).$

LF2I: Example – Type 1a Sne

- Consider the problem of fitting a cosmological model, e.g., $\Lambda\mathrm{CDM}$, to the Union 2.1[1] Type 1a supernova data set.

- The statistical model for this task is tractable (it is a 580-dimensional Gaussian), but the problem provides a simple benchmark to illustrate the LF2I algorithm, and a recent variation[2].



1. https://www.supernova.lbl.gov/
2. A. Al Kadhim, HBP, O. F. Prosper, *Mach. Learn.: Sci. Technol.* **5** (2024) 015020

LF2I: Example – Type 1a Sne

We'll use the test statistic

$$\lambda(X,\theta) = \sum_{i=1}^{580} \frac{\left(x_i - \mu(z_i,\theta)\right)^2}{\sigma_i^2}$$

where $x_i \pm \sigma_i$ are the measured distance moduli, z_i the measured redshifts, θ the cosmological model parameters and $\mu(z_i,\theta)$ is the theoretical prediction for the distance modulus.

Our goal is to approximate $P(\lambda \leq \lambda_D|\theta)$.

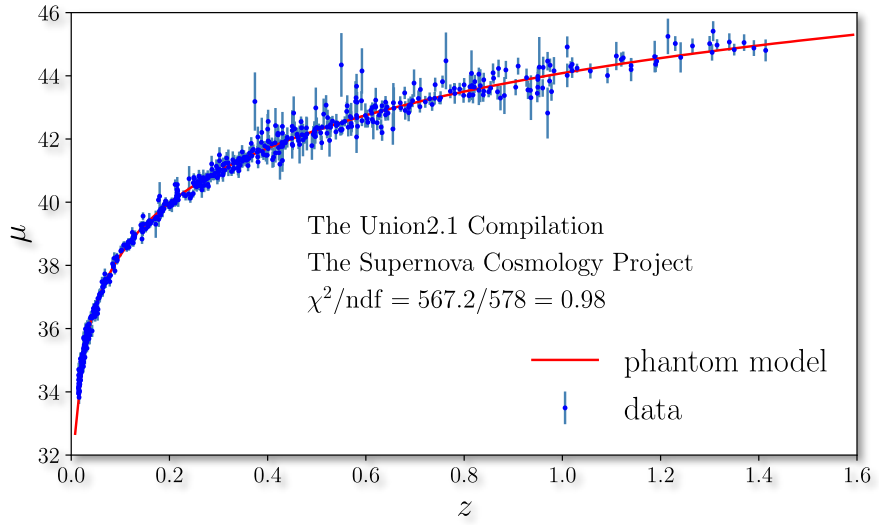
LF2I: Example – Type 1a Sne

We consider a cosmological model in which the energy density of the universe is given by

$$\Omega(a) = \frac{\exp(a^n - 1)}{a^3}$$

The model is defined by two parameters: the Hubble constant H_0 (not to be confused with the hypothesis symbol!) and the dimensionless parameter n.

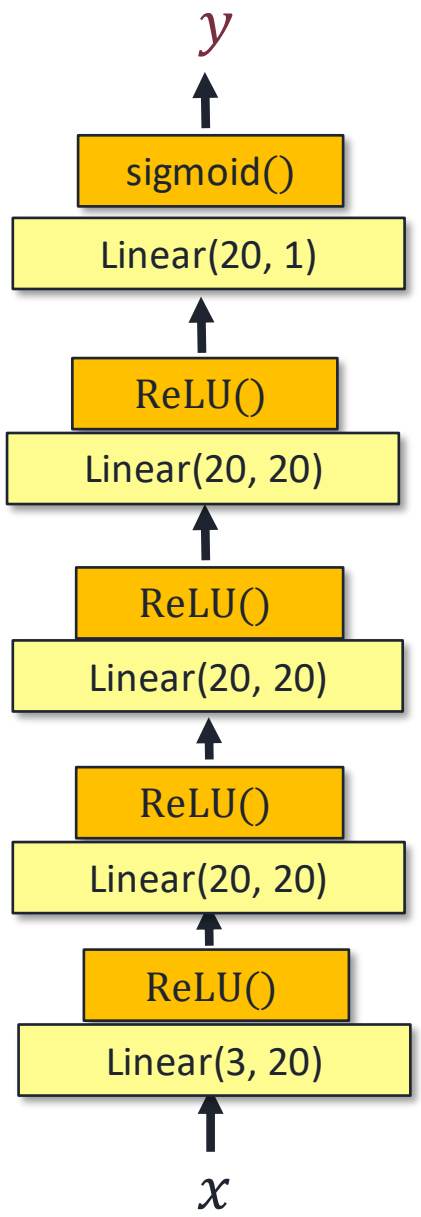
As the figure indicates, one gets an excellent fit of this model to the Type 1a data.



LF2I: Example – Type 1a Sne

Goal: model $y = P(\lambda \leq \lambda_D|\theta)$.[1]

- Inputs: $x = \lambda_D, n, H_0$
- Training data: 250,000 sets of 580 distance moduli sampled from Gaussians with the same standard deviations as the real data.
- Batch size: 50.
- Learning rate: 10^{-3}
- Adam optimizer.
- 100,000 iterations.



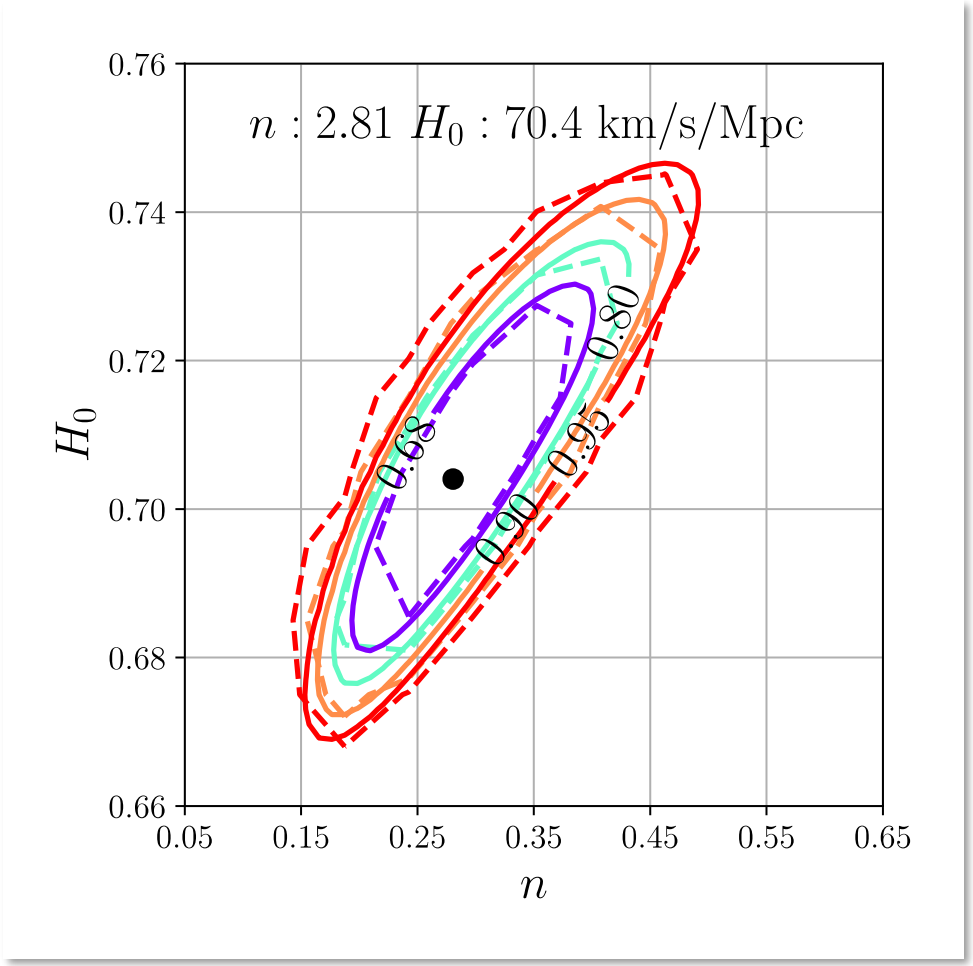
1. A. Al Kadhim, HBP, O. F. Prosper, *Mach. Learn.: Sci. Technol.* **5** (2024) 015020

LF2I: Example

Confidence set boundaries defined by $P(\lambda \leq \lambda_D|\theta) = 1 - \alpha$.

- Solid contours: NN.
- Dashed contours: histogram approximation of

$$\mathrm{E}[Z \mid \theta] = P(\lambda \leq \lambda_D|\theta).$$



Summary

- Simulation-based inference is a set of methods to make inferences about the parameters of a statistical model when the latter is intractable.

- The key ingredients are:

1. A prior density $\pi(\theta)$ for sampling the parameter space.
2. A simulator $\mathcal{F}(\theta)$ to simulate potential observations, X.
3. A test statistic $\lambda(X, \theta)$.
4. Machine learning models to approximate various statistical quantities given data sets of real and simulated data.